Chapter 13: Fiscal Policy, Deficits, and Debt

Practice Problems

Warm-Up: Quick Concept and Calculation Checks

a. To fight a recession, should the government use *expansionary* or *contractionary* fiscal policy? Name two specific tools it could use.

b. To fight demand-pull inflation, should the government use *expansionary* or *contractionary* fiscal policy?

c. True or False: automatic stabilizers require Congress to pass new legislation every time the economy changes.

d. As GDP rises in an expansion, what automatically happens to tax revenue and to transfer payments, without any new laws being passed?

e. A country's *actual* budget shows a deficit this year, but its *cyclically adjusted* budget is balanced. What does this combination suggest about whether the deficit was caused by fiscal policy choices or by the state of the economy?

f. What is the difference between *refinancing* the public debt and *retiring* it?

g. Why might economists consider debt as a *percentage of GDP* more meaningful than the absolute dollar amount of debt?

Problem 1: Closing a Recessionary Gap

An economy has a recessionary expenditure gap of $100 billion (aggregate expenditures at full-employment GDP fall $100 billion short of what is needed). The MPC in this economy is 0.8.

a. What is the multiplier?

b. How large an increase in government spending (G) alone would close this gap?

c. How large a tax cut (T) alone would close this gap? (Recall a tax change's effect is multiplied by MPC first, then by the multiplier.)

d. Propose one combination of a smaller increase in G together with a tax cut that, added together, would also close the gap exactly.

Problem 2: The Balanced-Budget Multiplier

The government increases spending by $30 billion and, in the same year, raises taxes by exactly $30 billion to keep the budget balanced. The MPC is 0.8.

a. Using the multiplier, what is the effect of the $30 billion increase in G alone on equilibrium GDP?

b. Using the tax multiplier ($MPC \times$ multiplier), what is the effect of the $30 billion increase in T alone on equilibrium GDP?

c. Combining (a) and (b), what is the net effect on equilibrium GDP?

d. Explain in one sentence why an equal-sized increase in G and T does not fully cancel out, even though the government's budget balance does not change at all.

Problem 3: Automatic Stabilizers, Progressive vs. Flat Taxes

Compare two hypothetical tax systems facing the same household, whose income falls from $60,000 to $50,000 during a downturn:

- **System A (progressive):** 10 percent on the first $40,000 of income, 25 percent on income above $40,000.
- **System B (flat):** 18 percent on all income, regardless of amount.

a. Compute this household's tax bill under System A at $60,000 income and at $50,000 income.

b. Compute the tax bill under System B at both income levels.

c. Under which system does the tax bill fall by *more* when income drops?

d. Compute disposable (after-tax) income under each system at both income levels, and the *drop* in disposable income under each. Which system cushions the household's spending power more?

Problem 4: Accumulating the Public Debt

A small country starts with zero public debt in Year 1. It then runs a deficit of $45 billion in Year 2, a surplus of $10 billion in Year 3, and a deficit of $20 billion in Year 4.

a. What is the country's public debt at the end of Year 4?

b. If this country's real GDP in Year 4 is $550 billion, what is its public debt as a percentage of GDP?

c. Suppose most of this debt is held by the country's own citizens and financial institutions (internally held) rather than by foreign lenders. Briefly explain why economists generally consider this less burdensome than the same debt held externally.

d. Name one way that financing this debt could *crowd out* private investment, connecting back to the market for loanable funds from Chapter 6.